

where we use the notation $\mathcal{K}^p := \mathcal{B}^p[-1]$ for all $p \in P$. Further, we call $\mathcal{K}$ the chain complex consisting of *preboundaries*.

Example 3.1. Consider the P -graded chain complex associated to the filtration in [Example 2.8](#). The chain group $\mathcal{C}_1^2$ admits a splitting $\mathcal{C}_1^2 \cong \mathcal{M}_1^2 \oplus \mathcal{K}_1^2$, where $\mathcal{M}_1^2$ is generated uniquely by $uv + vw$; see [Fig. 1](#). On the other hand, the preboundary $\mathcal{K}_1^2$ can either be chosen to be generated by uv or vw . As for $\mathcal{C}_1^1$, the single edge $uw \in \mathcal{C}_1^1$ generates a preboundary for the single boundary vertex $w \in \mathcal{C}_0^1$.

Throughout this section, we use the notation $\mathcal{A} := \bigoplus_{p \in P} \mathcal{A}^p$, for $\mathcal{A} \in \{\mathcal{M}, \mathcal{B}, \mathcal{K}\}$. In addition, we fix an isomorphism $\zeta: \mathcal{C} \cong \mathcal{M} \oplus \mathcal{B} \oplus \mathcal{K}$ such that, for each $p \in P$ and $n \in \mathbb{Z}$, it restricts to a splitting isomorphism $\zeta_n^p: \mathcal{C}_n^p \cong \mathcal{M}_n^p \oplus \mathcal{B}_n^p \oplus \mathcal{K}_n^p$. Also, we abuse notation and use ‘the submodules $\mathcal{M}$, $\mathcal{B}$ and $\mathcal{K}$ of $\mathcal{C}$ ’, to refer to the actual submodules $\zeta^{-1}(\mathcal{M})$, $\zeta^{-1}(\mathcal{B})$ and $\zeta^{-1}(\mathcal{K})$, respectively. In particular, given a chain $c \in \mathcal{C}$, there are unique chains $z \in \mathcal{M}$, $b \in \mathcal{B}$ and $w \in \mathcal{K}$ such that $c = z + b + w$. We also consider the inclusions and projections $\iota_{\mathcal{A}}: \mathcal{A} \hookrightarrow \mathcal{C}$ and $j_{\mathcal{A}}: \mathcal{C} \twoheadrightarrow \mathcal{A}$ for $\mathcal{A} \in \{\mathcal{M}, \mathcal{K}, \mathcal{B}\}$.

Before moving on, we need to add another condition on ζ . Namely, we assume that $d_n(\mathcal{K}_n^p) \subseteq \mathcal{B}_{n-1}^p$ for all $n \in \mathbb{Z}$ and $p \in P$ and say that ζ is *consistent*. Further, in this case, notice that d_n sends $\mathcal{K}_n^p$ isomorphically to $\mathcal{B}_{n-1}^p$. Later, in [Proposition 4.1](#), we show that such a splitting exists and that can be easily computed.

One can formulate the multicomplex $\mathcal{C}$ as a based complex where, for each degree n , there is an isomorphism

$$\mathcal{C}_n \cong \left(\bigoplus_{p \in P} \mathcal{M}_n^p \right) \oplus \left(\bigoplus_{p \in P} \mathcal{B}_n^p \right) \oplus \left(\bigoplus_{p \in P} \mathcal{K}_n^p \right) = \bigoplus_{\mathcal{A}_n^p \in I_n} \mathcal{A}_n^p$$

where we have used $I_n = \{\mathcal{A}_n^p \mid \mathcal{A} \in \{\mathcal{M}, \mathcal{K}, \mathcal{B}\}, p \in P, \mathcal{A}_n^p \neq 0\}$. Thus, given $p \leq q$ from P , the twisted differential $t^{pq}: \mathcal{C}^q \rightarrow \mathcal{C}^p$ can be written as $t^{pq} = \sum_{\mathcal{A}, \mathcal{D} \in \{\mathcal{M}, \mathcal{B}, \mathcal{K}\}} t_{\mathcal{AD}}^{pq}$, where the summands are given by $t_{\mathcal{AD}}^{pq} = \iota_{\mathcal{A}} \circ t^{pq} \circ j_{\mathcal{D}}$. Further, using that ζ is a consistent splitting, it follows that $t^p = t_{\mathcal{BK}}^p$ for all $p \in P$ since all other components are zero.

Next, we proceed to apply algebraic Morse theory as presented in [Section 2.4](#). To start, we consider the associated digraph $G_{\mathcal{C}}$ with vertex set $\bigcup_{n \in \mathbb{Z}} I_n$, together with the arrows $\mathcal{D}_n^q \rightarrow \mathcal{A}_{n-1}^p$ for $(t_{\mathcal{AD}}^{pq})_n \neq 0$ and $p \leq q$ from P , $n \in \mathbb{Z}$ and $\mathcal{A}, \mathcal{D} \in \{\mathcal{M}, \mathcal{B}, \mathcal{K}\}$.

We define a partial matching μ on $G_{\mathcal{C}}$ which contains all arrows of the form $\mathcal{K}_n^p \rightarrow \mathcal{B}_{n-1}^p$ which are in $G_{\mathcal{C}}$. In fact, notice that μ is a Morse matching. To see this, first, since ζ is consistent, we know that $t_{\mathcal{BK}}^p: \mathcal{K}_n^p \rightarrow \mathcal{B}_{n-1}^p$ is an isomorphism for all $p \in P$ and all $n \in \mathbb{Z}$. On the other hand, since we assume that P is finite, it follows that $G_{\mathcal{C}}$ is also finite. Also, one can see that there are no directed cycles in $G_{\mathcal{C}}^{\mu}$. This is because, by definition, all arrows from $G_{\mathcal{C}}^{\mu}$ are of the form $\mathcal{D}_n^q \rightarrow \mathcal{A}_{n-1}^p$ and such that $p \leq q$, and, if $p = q$, the only possible arrow is $\mathcal{B}_n^p \rightarrow \mathcal{K}_{n-1}^p$. Altogether, using [Lemma 2.12](#), it follows that μ is a Morse matching. In particular, notice that $\mu_n^0 = \{\mathcal{M}_n^p \mid p \in P \text{ and } \mathcal{M}_n^p \neq 0\}$.

Hence, we are equipped with the based complex $\mathcal{C}$ together with a Morse matching μ on $G_{\mathcal{C}}$, and so we can follow the exposure from [Section 2.4](#) to obtain a specialised result. During the rest of this section, we follow such strategy.

To start, we define a based complex $\tilde{\mathcal{C}}$ which is equal to $\mathcal{C}$, except for the differential $\tilde{d}$, which is $\tilde{d} = \iota_{\mathcal{B}}(\sum_{p \in P} t_{\mathcal{BK}}^p)j_{\mathcal{K}}$. In particular, notice that $\tilde{d}$ is a P -filtered map. On the other hand, we define the based complex $\tilde{\mathcal{M}}$ which is given by $\tilde{\mathcal{M}}_n = \bigoplus_{p \in P} \mathcal{M}_n^p$ together with the trivial differential. As in [Section 2.4](#), using [[Skö18](#), Lemma 1], there is a P -filtered contraction

$$\tilde{\mathcal{M}} \xleftarrow[\tilde{g}]{\tilde{f}} \tilde{\mathcal{C}} \rightrightarrows \tilde{h}$$

where (one can check that) $\tilde{f}: \tilde{\mathcal{C}} \rightarrow \tilde{\mathcal{M}}$ and $\tilde{g}: \tilde{\mathcal{M}} \rightarrow \tilde{\mathcal{C}}$ are given by $\tilde{f} = j_{\mathcal{M}}$ and $\tilde{g} = \iota_{\mathcal{M}}$, respectively, while $\tilde{h}: \tilde{\mathcal{C}} \rightarrow \tilde{\mathcal{C}}[1]$ is given by $\tilde{h} = -\iota_{\mathcal{K}}(\sum_{p \in P} (t_{\mathcal{BK}}^p)^{-1})j_{\mathcal{B}}$. In fact, it is not difficult